

INVESTIGATEIQ

Product Blueprint

Evidence-to-decision workflow for fictional AML investigations

Blueprint purpose

This document explains what InvestigateIQ does, how a user moves through the product, how evidence and retrieval are represented, and where human accountability remains required. It is a product and technical blueprint for a fictional teaching environment—not a deployment or regulatory filing design.

Prepared for Capstone Group 7 · Leadership with AI, IIT Bombay
Version 1.2 · 10 October 2026

Document guide

The blueprint is organized for a reviewer who wants both the product story and the implementation logic.

Section	What it answers
1. Product intent	What problem the product demonstrates and what it deliberately does not claim.
2. User journey	How one fictional case moves from the queue to a human decision and, if needed, Compliance.
3. Screen blueprint	What each current page, tab, accordion, and action is for.
4. Evidence and RAG	How source records, chunks, keyword search, citations, and the confidence score work.
5. Technical design	The Streamlit router, SQLite stores, major calculations, safeguards, and deployment model.
6. Implementation logic	The actual flow for OKF ingestion, chunking, FTS5/BM25 retrieval, RAG citation checks, confidence, and audit states.
7. Responsible AI	Why the system assists review but never decides, closes, escalates, or files automatically.
8. FAQ and demo preparation	Detailed answers for faculty questions plus a safe two-case demonstration sequence.

Scope note

- All customers, alerts, transactions, documents, and source material are fictional course data.
- The product does not authenticate real users, connect to a bank, verify outside-bank records, or file a report with a regulator.
- A record, citation, retrieval result, rule match, or confidence score is an aid to review—not proof that money is lawful, unlawful, fraudulent, or laundered.

Lecturer submission alignment

Selected submission track: Blueprint + Prototype + Detailed code + Final results. The submission includes this detailed blueprint, the working prototype and live walkthrough, the documented Streamlit/SQLite/OKF/RAG implementation logic, current screen evidence, validation results, and the 15-minute presentation materials.

Requirement	How this submission addresses it
Detailed blueprint	Documents the program statement, user workflow, evidence model, coding logic, responsible-AI limits, ROI measurement plan, and future roadmap.
Prototype and detailed code	The working Streamlit prototype is demonstrated live; the Blueprint maps its router, page modules, SQLite stores, OKF ingestion, chunking, FTS5/BM25 retrieval, confidence checks, and audit-state logic.
Final results	Provides current product captures, validation/deployment evidence, a rebuilt walkthrough plan, and measurable pilot success criteria rather than unsupported performance claims.
15-minute presentation	Soumya is the presenter and Laxman is the backup. The concise core deck is timed for 5 minutes 35 seconds, followed by an approximately five-minute live demonstration. Questions are handled within the same 15-minute session; the three FAQ slides are an appendix used only when asked.

Project contributors

Contributor	Project focus
Rahul Chainani	Application development and integration.
Pankaj Bharsakale	AI workflow and evidence review.
Lakshmi Dudgikar	Requirements and case-study mapping.
Rahul Sharma	Quality assurance and edge cases.
Laxman Singh	User experience and product walkthrough.
Soumya	Team coordination and presentation.
Sweta Singh	Course linkage and project explanation.

1. Product intent and design principles

Program statement

InvestigateIQ is a capstone program to design and demonstrate a transparent, AI-assisted workflow for fictional AML case review. The program connects a stored alert to inspectable case material, source chunks, bounded assistance, evidence-confidence limitations, a named human rationale, and a separate Compliance review. Its success is measured by traceability, review clarity, and responsible human oversight—not by autonomous decisions, legal conclusions, or external filing.

Program element	Defined commitment
Problem	A monitoring alert alone does not show the supporting material, missing evidence, or accountable next step.
Primary users	Investigators, Team Leads, Compliance Officers, and Admin reviewers in a fictional learning environment.
Delivered capability	Queue, workspace, evidence/RAG, bounded Copilot, confidence signal, audit trail, Compliance outcomes, analytics, and protected Admin inspection.
Success measures	A reviewer can trace a report claim to stored material, identify evidence gaps, distinguish workflow states, and record a rationale without an automated decision.
Non-goals	No real customer data, real authentication, autonomous escalation/closure/referral, legal finding, or regulatory filing.

InvestigateIQ is an AI-assisted case-investigation copilot. It turns a stored fictional alert into a transparent review workspace: the investigator can inspect the underlying case material, ask a bounded question, trace the source text behind a response, and then record an accountable human action.

Design principle	Implementation consequence
Traceability	Evidence & RAG exposes the source record, exact indexed chunk, retrieval method, and stated evidence gaps.
Human accountability	Only a named investigator or Compliance Officer can submit an action, and a written rationale is required.
Bounded assistance	The Copilot answers from stored fictional rows and chunks. It cannot independently verify documents or take workflow actions.
Clear states	Open, escalated, source-closed, simulated closure, Compliance return, acknowledged, and referred outcomes are kept distinct.
Responsible presentation	Evidence confidence measures support coverage for a draft, not risk, guilt, legality, or a recommended case outcome.

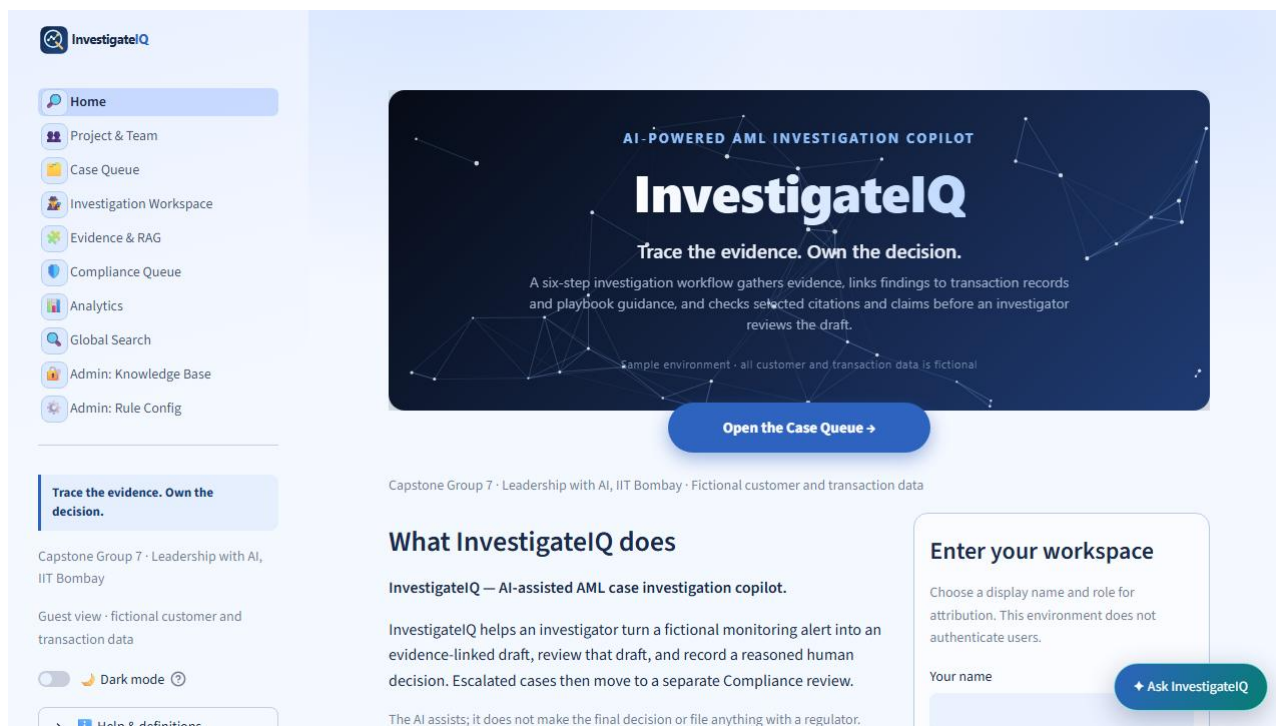


Figure — Current Home screen and reordered navigation. The workspace-entry panel establishes a display identity for the fictional session.

2. End-to-end user journey

A complete review follows the path below. The system preserves the difference between an alert, a drafted explanation, supporting material, a human choice, and a second human review.

Step	User action	What the system provides	Boundary
1. Find work	Use Case Queue filters or Global Search.	Stored alert metadata, status, severity, rule labels.	An alert is not a finding.
2. Review case	Open Investigation Workspace and choose saved or calculated mode.	Case context, report, validator notes, confidence, cited material.	The AI/report does not submit a decision.
3. Inspect proof	Open Evidence & RAG and exact chunks.	Case records, source guidance, chunks, retrieval results, method notes.	Citations point to supplied text; they do not authenticate it.
4. Decide	Record request-for-information or escalation with a rationale.	Locked recorded-action state and append-only audit history.	Only a human can act; completed actions cannot be edited in place.
5. Second review	A Compliance Officer acknowledges, returns, or records an onward referral.	Outcome mapping back to the queue and audit history.	The tool never files externally.

Case Queue: the operational starting point

Queue cards summarize the fictional workload. Filters narrow by severity, status, scenario, and customer name. A row's status explains where it is in the workflow; its action opens the appropriate context. Completed cases expose Evidence rather than an invitation to alter a submitted decision.

InvestigateIQ

Home
Project & Team
Case Queue
Investigation Workspace
Evidence & RAG
Compliance Queue
Analytics
Global Search
Admin: Knowledge Base
Admin: Rule Config

Trace the evidence. Own the decision.

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Demo Investigator
Role: Investigator
Dark mode

Source closed
2
Evidence not established

High severity
17
Source priority label

Only recorded human actions count as Human closed. Source-closed cases have not been independently substantiated here. High severity is a separate source label and can overlap any status. Hover over a card or open the definitions below.

What these queue numbers mean

Severity Status Scenario Search customer name
Choose options Choose options Choose options e.g. Apex

Showing 47 of 47 alerts

How Compliance outcomes appear in the Case Queue

Case	Customer	Alert Type	Severity	Status	Rule(s)	Action
FIC-CASE-A81E9BDDF8A D	Generated fictional customer	Calculated fictional review signal	Review	Open	Calculated: 24h / 5x / 80% / 3...	Ask InvestigateIQ
FIC-CASE-47F85A32F6E 3	Generated fictional customer	Calculated fictional review signal	Review	Open	Calculated: 24h / 5x / 80% / 3...	Ask InvestigateIQ
CASE-043	Arjun Mehta	Rapid incoming funds	High	Open	Calculated: 24h	Ask InvestigateIQ

Ask InvestigateIQ

Figure — Queue filters and case table. The search field, current status, rule metadata, and action are separate controls.

3. Investigation Workspace

The Investigation Workspace is the main review surface. A case selector and mode control make the source of the displayed report explicit.

Mode	Purpose	Safety boundary
Saved report	Replays a historical saved snapshot for a repeatable demonstration.	It must not be confused with current index contents or a fresh model call.
Calculated	Recomputes the review from current stored fictional rows without a model call.	A calculation still depends on supplied data and does not verify external facts.
AI draft	Optional live drafting mode when configured.	A draft is subject to evidence checks and remains advisory; a human action is still required.

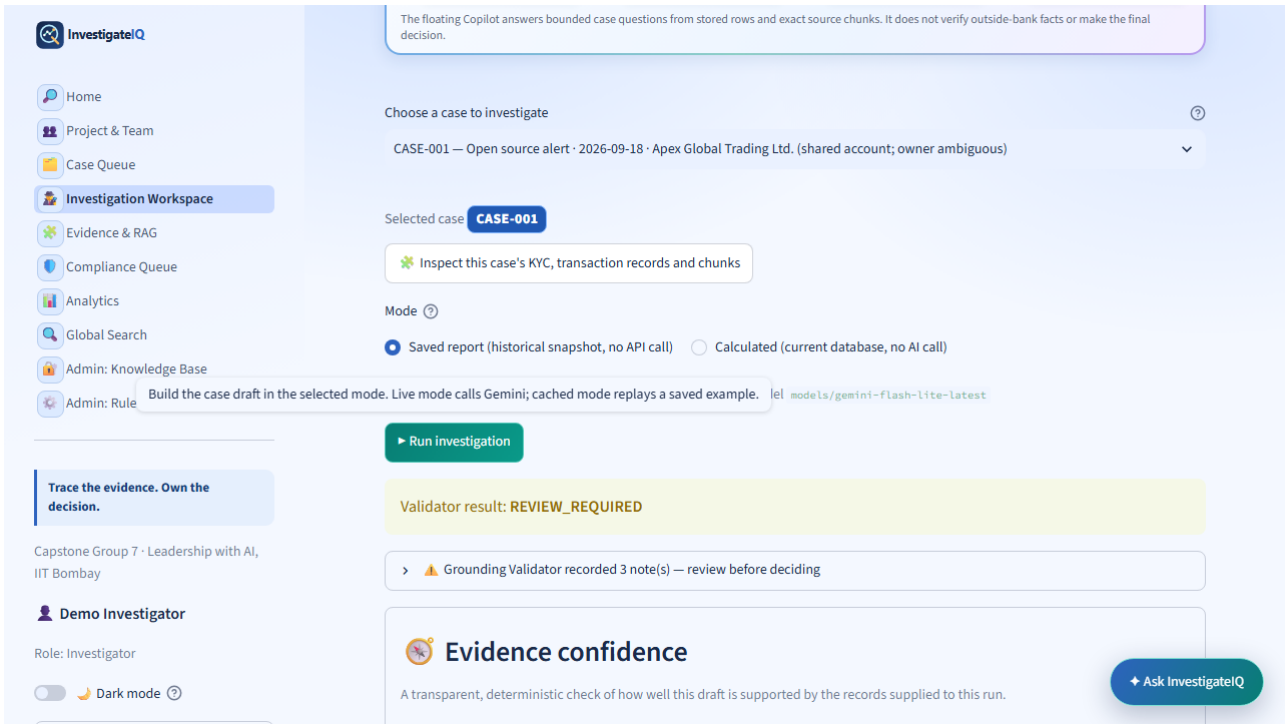


Figure — Workspace report controls and validator result. Run investigation builds or replays a review; it does not record a decision.

Evidence confidence

The evidence-confidence card is the visible responsible-AI control. It brings uncertainty into the case report instead of hiding it behind a confident sounding narrative.

- Evidence support score: deterministic summary of how much of the draft is supported by the supplied evidence and guidance.
- Evidence support: an interpretable label such as Low, paired with the count of known limitations.
- Explicit prohibition: it is not a probability of crime, money laundering, legality, or the correct workflow outcome.

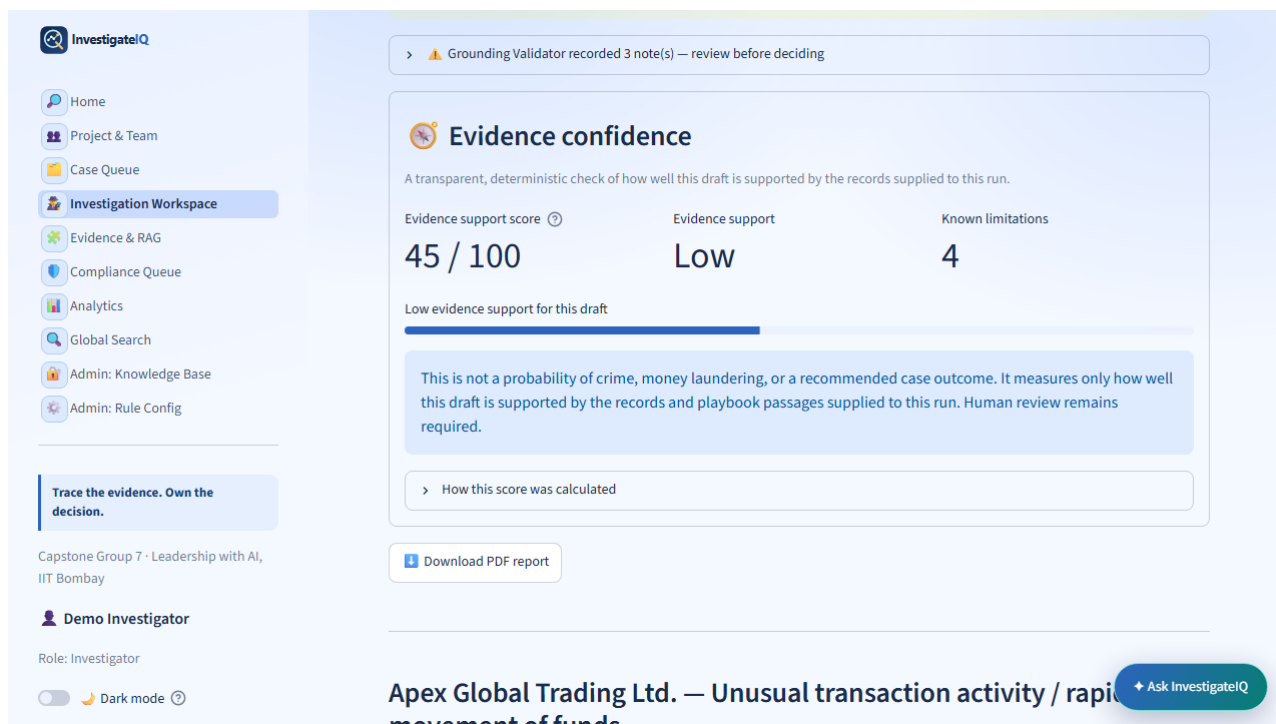


Figure — Evidence confidence is shown after report generation, with score, label, limitations, and a visible human-review requirement.

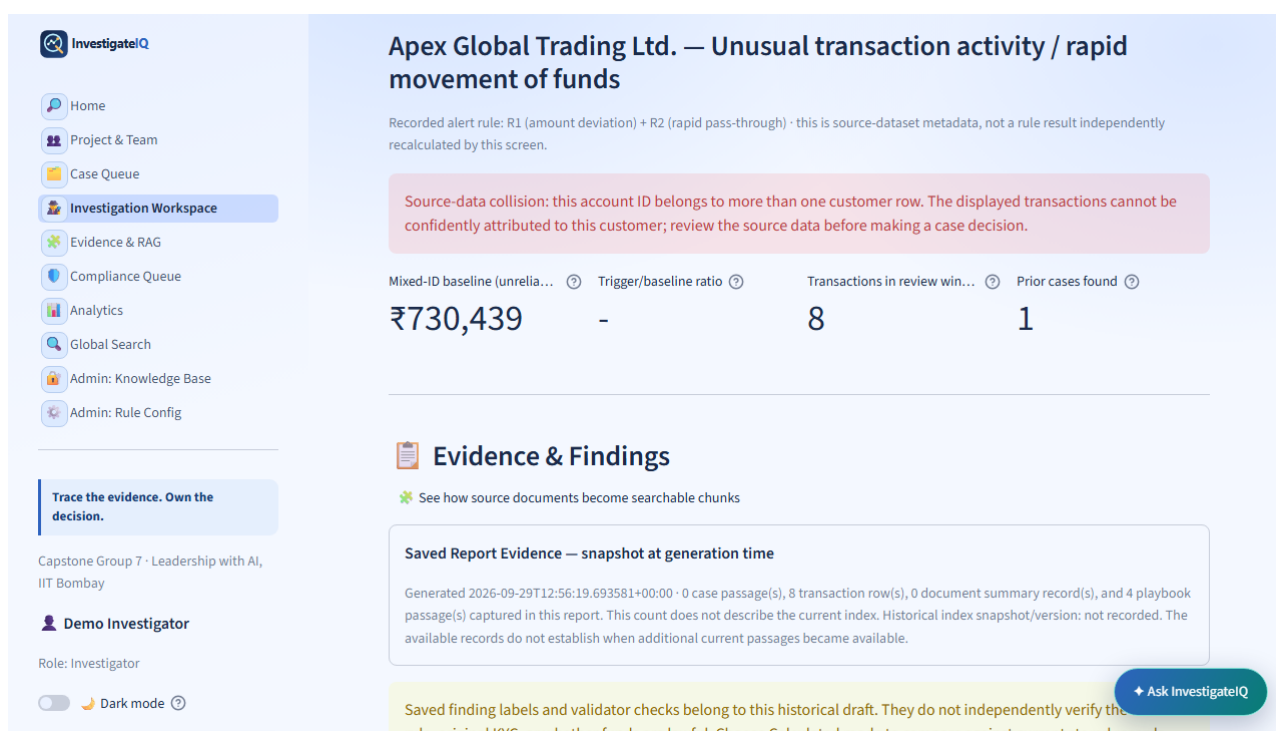


Figure — Case context and Evidence & Findings retain source-data warnings alongside the report material.

4. Evidence & RAG architecture

Evidence & RAG makes the source trail inspectable. The product uses retrieval-augmented generation in a narrow, transparent form: it finds relevant stored passages first, then lets the user inspect the passages that informed a draft or answer.

Tab	Primary purpose	Reviewer takeaway
Case records	Inspect the selected case's records, evidence gaps, indexed passages, and transaction scope.	Know what is present, what is illustrative, and what is missing.
Rule Lab	Explain calculated signals and preview logic.	A rule match calls for review; it is not a verdict.
Fictional Intake	Show the separate generated packet and its calculated evidence boundary.	Generated data is labeled separately from imported alert data.
Source & chunks	Open source guidance and exact chunk boundaries.	A citation can be traced to exact text.
Search the index	Run a keyword query against stored chunks.	Search results are inspectable leads, not automatically true answers.
Method & code	Explain ingestion, chunking, retrieval, context gathering, and checks.	The implementation is visible and its limits are stated.

The screenshot displays the InvestigateIQ interface. On the left is a sidebar with navigation links: Home, Project & Team, Case Queue, Investigation Workspace, **Evidence & RAG** (selected), Compliance Queue, Analytics, Global Search, Admin: Knowledge Base, and Admin: Rule Config. Below the sidebar, user information is shown: Capstone Group 7 - Leadership with AI, IIT Bombay, and a Demo Investigator role.

The main content area is titled "Evidence & RAG" with the subtitle "Inspect fictional case records, their exact chunks, and separately indexed investigation guidance." Below this, it states "Sample environment - all customer and transaction data is fictional".

A section titled "How this page works" explains the purpose: "Follow a record from source to review". It outlines a four-step process:

- STEP 01 Choose a case**: See the customer profile, illustrative identity records, alert and transaction packet.
- STEP 02 Inspect source and chunks**: Compare complete generated text with each exact indexed paragraph.
- STEP 03 Check coverage**: See what is missing, including original documents and independent verification.
- STEP 04 Test and search**: Recalculate a fictional alert, then inspect the separate playbook chunks.

Below the steps, a note states: "Source-database investigations retrieve case passages and separate playbook guidance through SQLite FTS5; older saved reports may predate the case index. Fictional Intake has its own case-scoped chunk search and calculated review. No vector embeddings are used, and retrieval does not establish lawful funds."

Statistics are provided: "24 synthetic source documents · 57 indexed chunks · SQLite FTS5 keyword retrieval".

A key insight is highlighted: "A saved investigation cites a source document ID. That identifies the source playbook, but does not by itself prove which individual chunk supported every sentence. Review the text and original records before relying on a draft."

At the bottom, the "Evidence exploration workspace" is shown with the instruction: "Trace the source record, exact indexed chunk, and retrieval method behind a finding." There is an "Explore evidence" link and a button labeled "Ask InvestigateIQ".

Figure — Evidence & RAG overview and its four-step evidence-review flow.

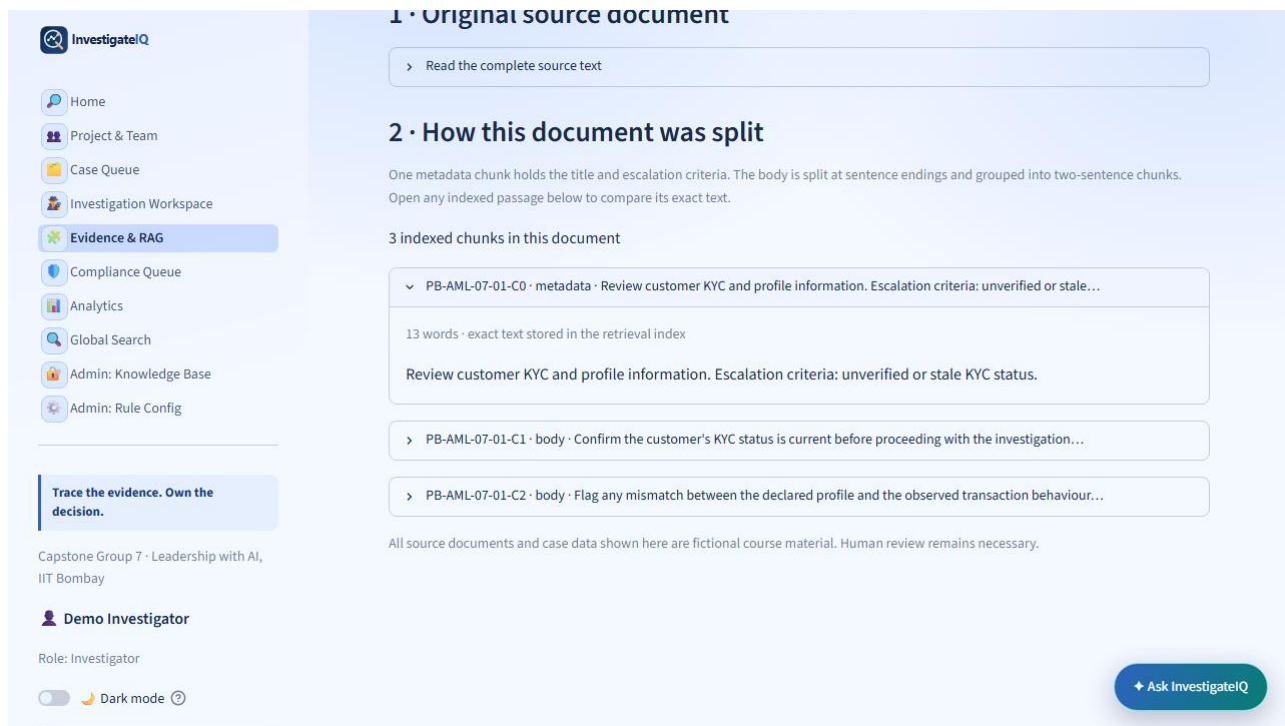


Figure — Source & chunks shows exact stored chunk boundaries and labels instead of only an AI summary.

Retrieval logic in plain language

Guidance documents are stored as synthetic source records. The builder creates metadata chunks and body chunks, indexes their text in SQLite FTS5, and uses BM25 ranking for a keyword query. Case records are also indexed in their own scope. A retrieved identifier can be checked against the selected source set before a cited report element is accepted.

```
source document → metadata/body chunks → SQLite FTS5 index
case scenario + question → scoped keyword search → BM25-ranked passages
retrieved passage IDs → draft/answer citations → human opens exact text
```

5. The Copilot and human decision boundary

The floating InvestigateIQ Copilot gives a page-aware entry point to bounded case questions. It is intentionally separate from the decision form: a useful answer is not the same thing as a workflow action.

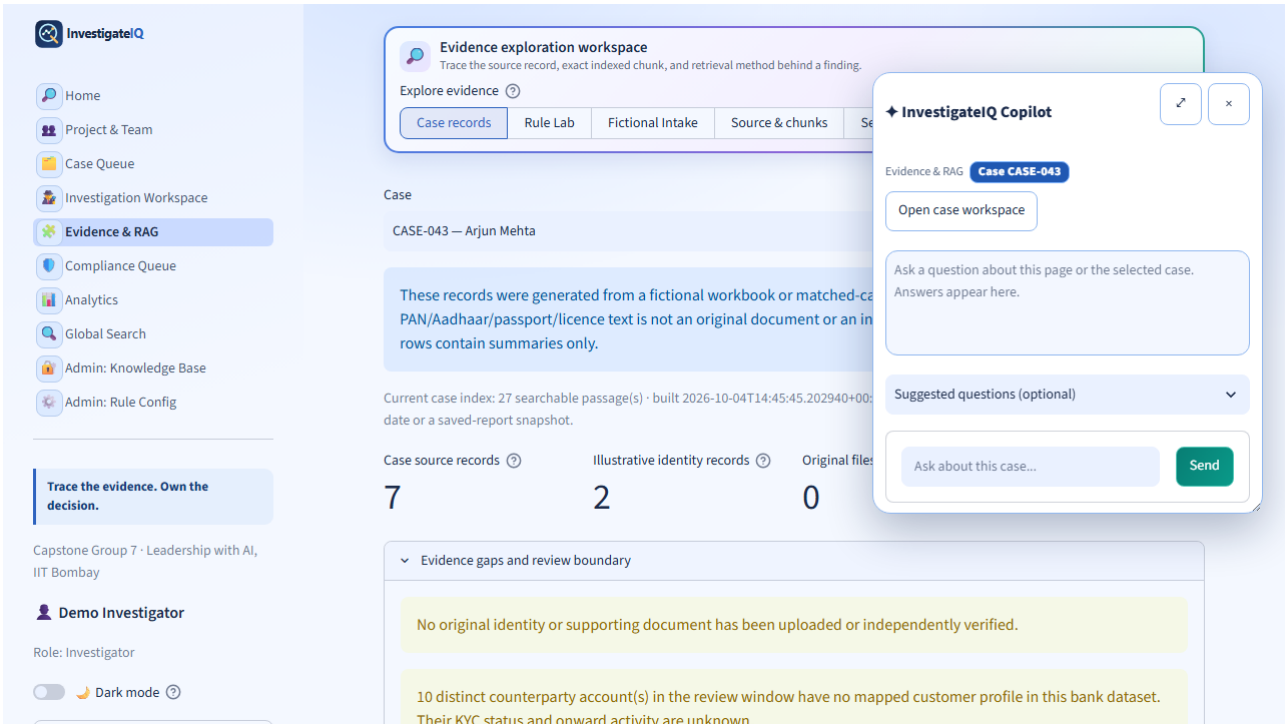


Figure — The floating Copilot opens over the current page while retaining the selected case context.

Can do	Cannot do
Summarize stored fictional records and retrieved source chunks.	Verify a source document, counterparty, KYC record, or outside-bank fact.
Point the user to exact IDs, chunks, and evidence gaps.	Conclude that funds are lawful, unlawful, fraudulent, or laundered.
Help the user frame a question or understand a result.	Close, escalate, refer, submit, or file anything on the user’s behalf.

Human decision logic

Decision controls are enabled only while the case has no final investigator action, or after Compliance returns the case for further information. On submission, the action, investigator display name, rationale, and timestamp are appended to the audit history and the form is replaced by its recorded state.

```
if latest_action is None or latest_action == 'compliance_return':
    require(investigator_name and rationale)
    append_human_action(case_id, selected_action, rationale)
else: show_recorded_action_and_disable_form()
```

Why the form locks

A human decision is an accountable audit event, not a chat preference. Locking the completed form prevents a user from silently changing a recorded decision. A new permitted workflow state, such as a Compliance return, creates a new auditable action rather than overwriting the original.

6. Compliance outcomes and state mapping

Compliance is a separate human review of investigator escalations. Its three actions have different operational meaning and need distinct queue labels.

Compliance action	Queue indication	Meaning
Acknowledge — no further action	Compliance reviewed — no further action	The escalation was reviewed; no additional action is needed in this fictional workflow.
Return to investigator	Compliance returned — information needed	The case goes back for more evidence or explanation before Compliance can decide.
Referred onward	Compliance referred onward — recorded	A human recorded an onward referral. The tool itself does not file anything with a regulator.

InvestigateIQ

- Home
- Project & Team
- Case Queue
- Investigation Workspace
- Evidence & RAG
- Compliance Queue**
- Analytics
- Global Search
- Admin: Knowledge Base
- Admin: Rule Config

Trace the evidence. Own the decision.

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Demo Investigator

Role: Investigator

Dark mode

Compliance Queue Workflow:

- STEP 01: Open an escalation** — See the case and the investigator's recorded rationale.
- STEP 02: Review the context** — Read the rationale; open the Workspace if you need the evidence.
- STEP 03: Record an action** — Acknowledge, return, or record a referral in the audit trail.

A recorded referral is not a regulatory filing; a human officer remains responsible.

Your selected role is **Investigator**. This screen is normally used by **Compliance Officer**. Access is not enforced here; any action will be attributed to your selected display identity.

Cases awaiting Compliance review: **0**

High-severity among them: **0**

No cases are currently escalated. Escalate a case from the Investigation Workspace to see it here.

Scope note: this screen never drafts or files a regulatory report — that decision and action stay with Compliance/MLRO outside this tool, per the documented out-of-scope boundary. All data is fictional.

[Ask InvestigateIQ](#)

Figure — Compliance Queue makes the separate second-review boundary visible.

7. Analytics and Global Search

Analytics is descriptive and uses carefully named axes. The graphs show counts and proportions in the stored fictional dataset; they are not model-quality statistics or probability estimates.

Graph	X-axis / measure	Interpretation guardrail
Alerts by severity	Number of stored alerts	Counts High, Medium, and Low labels.
Alerts by typology	Number of stored alerts	Counts categories such as velocity or structuring.
Alerts by current status	Number of cases	Counts current workflow states; source status and human state are distinct.
High-severity share	Percentage	Share labeled High within a typology; not probability of crime.
Stored records by month	Number of alerts per time bucket	Describes fictional source dates; not a risk forecast.
Recorded decisions/actions	Saved action count	Counts workflow events; not productivity, accuracy, or outcome quality.



Figure — Analytics graphs with readable axes and explanatory units.

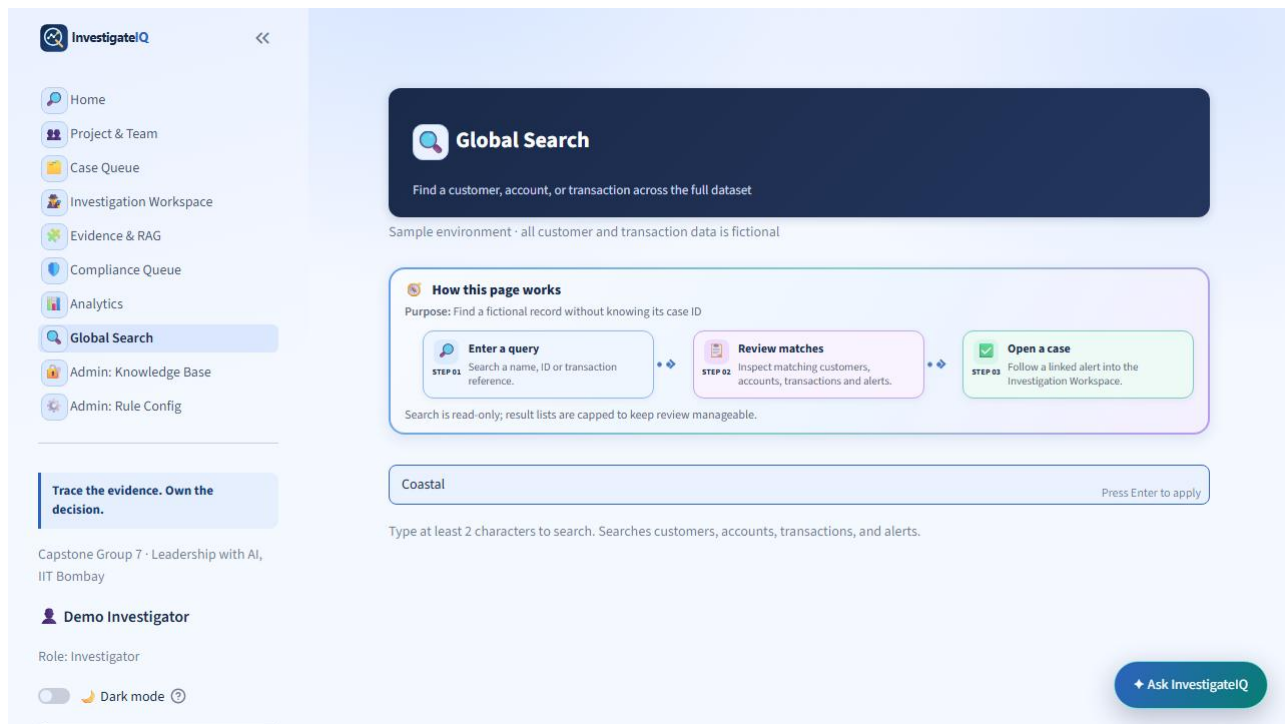


Figure — Global Search returns linked fictional customers, accounts, transactions, and cases as a context-finding tool.

8. Project roles and protected administration

Project & Team makes responsibilities visible and keeps names separate from email addresses. The selected UI role is a display/workflow label; it is not identity authentication or an authorization grant.

customer_id	name	type	occupation_or_industry	risk_rating	kyc_status
CUST-0446	Coastal Impex Ltd.	Business	Logistics	High	Verified
CUST-0454	Coastal Logistics LLP	Business	IT services	Low	Verified
CUST-0485	Coastal Suppliers Ltd.	Business	IT services	Low	Verified
CUST-0503	Coastal Impex Pvt. Ltd.	Business	Import/export	Medium	Verified
CUST-0518	Coastal Industries Ltd.	Business	Wholesale trade	Medium	Verified
CUST-0521	Coastal Ventures Ltd.	Business	Logistics	Low	Verified
CUST-0544	Coastal Impex Pvt. Ltd.	Business	Manufacturing	High	Verified
CUST-1005	Coastal Wholesale Traders	Business	Wholesale distribution - consu	Medium	Verified

account_id	customer_id	customer_name	type	status	risk_rating
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Figure — The role guide: Investigator, Team Lead, Compliance Officer, and Admin.

Admin boundaries

Knowledge Base and Rule Config are gated by a separate masked passcode. The passcode is never included in this Blueprint or automated capture. The locked screen makes the boundary apparent without turning a coursework environment into an authentication claim.

Admin — Knowledge Base Inspector

Inspect the retrieval sources used to support investigation drafts.

Sample environment · all customer and transaction data is fictional

How this page works

Purpose: Show what guidance the copilot can retrieve

STEP 01: Unlock Enter the admin passcode.

STEP 02: Inspect View fictional playbooks, document chunks and index contents.

STEP 03: Search Try the keyword retrieval used to ground report drafts.

This is a synthetic teaching knowledge base, not an approved bank or RBI policy repository.

Your selected role is **Investigator**. This screen is normally used by **Admin**. Access is not enforced here; any action will be attributed to your selected display identity.

This passcode is a limited access gate, not production-grade authentication or authorization.

Admin passcode

Figure — Knowledge Base locked state. The protected source-inspection area remains visibly separate from ordinary workflow pages.

9. Technical architecture and code map

The current product is a Streamlit application with an explicit navigation router, page modules, shared UI helpers, SQLite-backed fictional stores, and static assets. The structure favors transparent page-level logic that can be inspected during a demonstration.

Layer	Representative responsibility	Examples
App router	Defines the visible order and labels of the left navigation.	Home; Project & Team; Case Queue; Investigation Workspace; Evidence & RAG; Compliance; Analytics; Search; Admin.
Page modules	Render the task-specific controls, tables, charts, and bounded workflow actions.	0_Case_Queue.py, 2_Investigation_Demo.py, 8_Evidence_RAG.py, 5_Analytics.py.
Shared UI	Banners, compact flows, status badges, help text, role behavior, theme styling, and the Copilot launcher.	ui_common helpers and shared CSS.
Data and search	Fictional case records, audit actions, documents, chunks, retrieval index, and calculated metrics.	SQLite, FTS5 keyword search, BM25 ranking, deterministic calculations.
Assets and release	Walkthrough media, screenshots, captions, deployment configuration, and reproducible builders.	assets/, scripts/, Procfile, GitHub main branch → Railway.

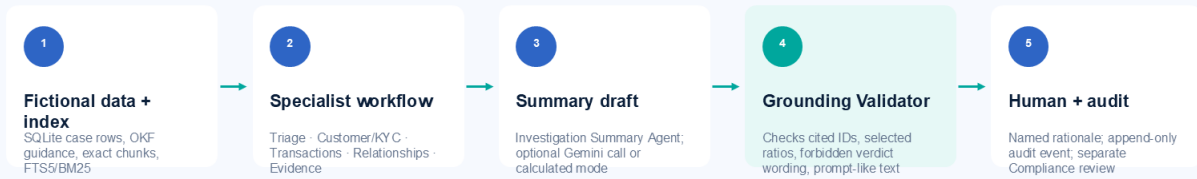
9.1 Detailed architecture: where technology and AI are used

The architecture intentionally separates evidence retrieval, deterministic calculations, optional drafting, validation, and human workflow. This makes it possible to show what is computed from stored fictional data, what may be drafted with a model, and what remains a human responsibility.

Stage	Actual component / logic	Output and control
1. Stored fictional data and index	SQLite stores the fictional case, alert, evidence, audit, OKF-source, and chunk records. FTS5 performs lexical keyword matching and BM25 ranks matching passages; no vector embedding index is used.	A case-scoped set of inspectable rows and chunks. Retrieval finds supplied text; it does not verify truth.
2. Evidence-gathering specialists	Alert Triage, Customer/KYC, Transaction Investigation, Relationship, and Evidence specialists read the active case and calculate or collect structured findings.	Structured evidence and limitation inputs. These are specialist components, not CEO/CFO persona agents.
3. Investigation Summary Agent	Combines the structured specialist outputs into a report. In optional model mode it can call Gemini; in calculated mode it creates the report without a model call.	A clearly labelled advisory draft, never an automatic case action or final outcome.
4. Grounding Validator	Checks known citation IDs, selected ratio claims, forbidden verdict wording, and obvious prompt-like text in supplied fields. It can downgrade unsupported verified language or remove unsafe output elements.	Visible validation notes and safer wording. It reduces selected failure modes; it cannot guarantee factual truth or completeness.
5. Human action and audit	A person supplies a display name and required rationale before a permitted decision is appended. Compliance can acknowledge, return, or record an onward referral as a separate human step.	Recorded workflow state and auditable history. The display name is not authenticated identity or production RBAC.

ARCHITECTURE: TECHNOLOGY, AI, AND GUARDRAILS

Six specialist components, one optional model draft, and a human decision



Where AI is used: the Investigation Summary Agent drafts a structured report only after the deterministic specialist components retrieve and calculate evidence. The Grounding Validator then checks selected claims before a human can act.

Important: these are investigation-specialist components—not CEO/CFO persona agents. The validator reduces specific errors; it cannot guarantee factual truth.

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Fictional AML investigation learning environment

Figure — Implemented architecture: deterministic specialist evidence gathering, optional Summary Agent drafting, Grounding Validator controls, and the final human/audit boundary.

9.2 Architecture execution path

The code-level path below is the most important explanation for a faculty reviewer: specialist outputs are assembled first; an optional model can draft a summary; a validator checks the draft against controlled rules; and only then does the UI present the material for a human decision.

```

triage → kyc → transactions → relationships → evidence → structured_inputs
raw_report = summary_agent(structured_inputs) # optional Gemini, or calculated mode
validated_report = grounding_validator(raw_report, known_ids, ratios, safe_text_rules)
show(validated_report, evidence_confidence, limitations); human_records_rationale()
  
```

Navigation router

```

st.navigation([Home, Project & Team, Case Queue, Investigation Workspace,
Evidence & RAG, Compliance Queue, Analytics, Global Search,
Admin: Knowledge Base, Admin: Rule Config]).run()
  
```

Confidence-score concept

The score is calculated after a report is available. It uses deterministic support and limitation checks over the draft, supplied evidence, and retrieved guidance. Its job is to surface coverage gaps before the decision—not to rank people or forecast misconduct.

```

confidence = calculate_evidence_confidence(report, evidence, guidance)
display(score, support_label, known_limitations, explanation)
never interpret confidence as crime_probability or recommended_outcome
  
```

10. Implementation logic: from source to auditable action

This section describes the product logic at code level. The intent is not to claim a production AML control; it is to make the demo's data flow, retrieval logic, and workflow safeguards inspectable.

10.1 Page routing and session context

app.py declares the left-navigation pages and runs the selected page module. The workspace entry page stores a display name and role in Streamlit session state so an audit entry can be attributed during the fictional demo. This is a UI/session convenience, not sign-in or authorization.

```
selected_page = st.navigation(page_definitions).run()
st.session_state['display_name'] = entered_name      # display attribution only
st.session_state['active_case_id'] = selected_case_id
```

10.2 Data model and case scoping

Store / entity	What is kept	How it is used
Case and alert rows	Fictional case ID, customer label, status, severity, rules, and scenario.	Drives Case Queue filters and determines the active case context.
Case evidence	Fictional customer, account, transaction, document, and evidence-gap rows.	Supplies the visible report and the supporting / missing-evidence checks.
Knowledge sources	OKF source metadata plus chunk text and chunk IDs.	Feeds Evidence & RAG and scoped keyword retrieval.
Audit actions	Named human action, rationale, timestamp, and preceding workflow state.	Creates the recorded decision state and Compliance history.

Every relevant query carries the active case ID or source scope. That prevents a case report from silently borrowing a different case's row or a citation from a non-retrieved source.

```
case_rows = load_case_rows(active_case_id)
guidance = search_chunks(question, scenario=case.scenario, case_id=active_case_id)
```

10.3 OKF ingestion and chunk construction

OKF means Open Knowledge Format in this product: a human-readable synthetic source record with stable metadata such as source ID, scenario, title, escalation criteria, version, and body text. The ingestion builder retains that metadata in a dedicated C0 chunk, then divides the body into reviewable groups of approximately two sentences. Chunk IDs preserve the parent source relationship, so a reviewer can move from a result back to the original source.

```
okf_source = {source_id, scenario, title, criteria, version, body}
chunks = [metadata_chunk(okf_source)] + two_sentence_chunks(okf_source.body)
save_chunk(chunk_id, source_id, scenario, chunk_text, chunk_type)
```

10.4 RAG retrieval, citation, and review

RAG means retrieval-augmented generation. InvestigateIQ uses a deliberately narrow implementation: it retrieves stored text first, ranks it with SQLite FTS5 and BM25, provides the resulting chunk IDs as context, and then displays citations that can be opened by a person. The product does not treat retrieval as verification, and it does not use a citation from outside the retrieved set.

```
matches = fts5_search(query_terms, scope=scenario_or_case)
ranked = bm25_rank(matches); context = top_chunks(ranked)
answer = build_bounded_answer(question, case_rows, context)
assert set(answer.cited_chunk_ids).issubset(set(context.chunk_ids))
```

Why FTS5 and BM25 are shown

FTS5 performs full-text keyword matching inside the local SQLite index. BM25 orders the matching passages by textual relevance. They help the user locate material; they do not measure truth, legal relevance, or the probability that an alert represents wrongdoing.

10.5 Report modes and evidence confidence

The Workspace labels whether a report was replayed from a saved snapshot, calculated from current fictional rows, or drafted by an optional AI configuration. After a report exists, deterministic support and limitation checks calculate the evidence-confidence card. The display deliberately pairs the score with support label, known limitations, explanation, and a human-review instruction.

```
report = load_saved_snapshot(case_id) # or calculate_from_rows(case_rows)
confidence = calculate_evidence_confidence(report, evidence, guidance)
show(score, support_label, limitations, 'Human review required')
```

10.6 Human and Compliance state machine

A decision is only persisted after the named person selects an allowed action and provides a rationale. The system appends a new audit event rather than editing an old one. The primary form then renders its recorded state. If Compliance returns the case, the next permitted investigator action becomes available as a new auditable step; the original action remains visible.

```
if decision_allowed(case_id): require(name and rationale); append_action(...)
else: render_recorded_state_and_disable_form()
if compliance_action == 'return': set_queue_state('information needed')
```

10.7 UI, tests, and release wiring

Shared UI helpers keep page banners, compact workflow flows, badges, tooltip treatments, theme colors, and the floating Copilot consistent. Local checks exercise the changed path before release; the repository's intended changes are committed and pushed; Railway serves the selected main-branch revision. The deployed product is then checked by visiting the affected page rather than relying only on a successful build log.

```
render_page_banner(...); render_compact_flow(...); render_tooltip(...)
local_test_changed_path() → git push main → Railway health/page verification
```

11. Testing, deployment, and operational checklist

The blueprint assumes local validation before release and a clear separation between read-only demonstration actions and persistent workflow actions.

Check	Expected result
UI state	Completed human decisions render as recorded and disabled; returned cases allow the next human action.
Evidence trace	A cited source or chunk opens to exact stored material with scope and limitation context.
Confidence	The score, support label, limitation count, and non-probability disclaimer appear together.
Accessibility/theme	Tooltip icons and tab text remain legible in both light and dark modes.
Analytics	Each graph has a clear title, axis/measure, and explanation of what it does not imply.
Release	Run the application locally, validate the changed path, commit only intended files, push main, and verify the deployed response.

Future enhancements

- Add repeatable test fixtures for every state shown in the Case Queue and Compliance outcome mapping.
- Add automated visual regression checks for dark mode, tooltips, responsive tables, and the full walkthrough captures.
- Version evidence-confidence rules and show a score-history explanation when a report is regenerated.
- Introduce real authentication and authorization only if the product moves beyond a fictional learning environment.

12. Value realization, ROI framework, and roadmap

The project does not claim a financial return from the fictional dataset. Instead, it provides a measurable ROI framework that a real sponsor could populate only after a governed pilot establishes baselines, costs, and observed reviewer outcomes.

ROI measurement framework

Value lever	Baseline and outcome to measure	Why InvestigateIQ may help
Review efficiency	Median minutes from opening an alert to locating the relevant case record and source passage.	Queue filters, Global Search, scoped records, and exact chunk links reduce navigation and lookup time.
Rework reduction	Number of Compliance returns or repeated evidence requests per completed review.	Visible gaps, cited support, and required rationale make missing information explicit before an action is recorded.
Audit readiness	Time to reconstruct the alert, cited text, decision, actor, rationale, and Compliance outcome for a sampled case.	Append-only audit events and source/chunk traceability preserve the review path.
Quality and safety	Rate of unsupported claims, broken citations, unclear state labels, or inaccessible controls found in review.	Bounded Copilot behavior, deterministic confidence limits, validators, and visual QA expose issues earlier.

```

annual_benefit = (cases × verified_minutes_saved × loaded_hourly_rate / 60)
                  + avoided_rework_cost + measurable_audit_preparation_savings
ROI % = ((annual_benefit - implementation_and_run_cost) / implementation_and_run_cost) × 100
payback_months = implementation_and_run_cost / monthly_verified_benefit

```

ROI guardrail

These formulas are a planning model, not a reported result. The program must collect a baseline, define the cost boundary, run a pilot, and verify the observed benefit before presenting any financial ROI figure.

Future roadmap

Horizon	Priority capability	Responsible delivery condition
Next iteration	Golden-question evaluation set, citation-quality tests, and additional repeatable fixtures for every queue and Compliance state.	Keep results tied to fictional test cases and publish failure cases alongside scores.
Pilot readiness	Role-based authentication, authorization, evidence-lineage versioning, and confidence-rule version history.	Use approved identity, privacy, and access-control requirements before any real data connection.
Scale with governance	Document ingestion controls, monitored retrieval evaluation, approved integrations, and reviewer-feedback analytics.	Establish data governance, human review protocols, model-risk assessment, and operational ownership first.

13. Detailed FAQ and final live-demo preparation

The following answers are prepared from the implemented demo, not from hypothetical features. They are intended for faculty questions during the same 15-minute presentation session. The FAQ appendix in the deck is deliberately short; this Blueprint retains the detailed version.

Likely question	Prepared answer
Where does the data come from?	All customers, cases, alerts, transactions, documents, and knowledge sources are fictional course material stored locally for this demonstration. The prototype is not connected to a bank feed, a live customer system, or an external regulator.
Why is a retrieved passage relevant?	The selected case and scenario scope the available material. SQLite FTS5 matches keywords in local text and BM25 ranks matching passages. The reviewer can open the source record or exact chunk. Relevance ranking is not truth verification.
Where is AI actually used?	Five specialist components gather and calculate evidence inputs. The Investigation Summary Agent may make the optional Gemini call to draft a structured report. Calculated mode makes no model call. The implementation does not use CEO/CFO persona agents.
How do you reduce hallucination or unsafe output?	The product uses supplied evidence and known source IDs, requires cited material, applies grounded wording rules, detects obvious prompt-like text, checks selected ratios and forbidden verdict language, and surfaces an evidence-confidence score with limitations. These controls reduce specific risks; they do not guarantee truth or eliminate hallucination.
Does a citation or confidence score prove that a case is legitimate or suspicious?	No. A citation is a pointer to supplied text and the confidence score is deterministic support coverage for a draft. Neither is a probability of crime, legality assessment, or recommended workflow outcome. A named human must interpret the evidence and record the decision.
Is the demo role selector real role-based access control?	No. It is a display/workflow label. Admin views use a limited passcode gate, but this is not production authentication or authorization. The passcode is not shown in the presentation, demo, Blueprint, or video.
What is recorded in the audit trail?	System, AI, and human workflow events retain the active case context. A human decision requires a name and written rationale and creates a new recorded state. Compliance review creates an additional auditable step rather than overwriting the earlier decision.

Likely question	Prepared answer
What is the business value or ROI?	The prototype does not claim savings from fictional data. A governed pilot would measure lookup time, avoided rework, audit-reconstruction time, and safety issues against an approved cost baseline before calculating financial ROI.

Who prepares which topic

Team member	Preparation topic	Live-session rule
Soumya	Primary presentation, product flow, and final delivery rehearsal.	Sole presenter unless unavailable.
Laxman Singh	Backup rehearsal, navigation, and demo safety checks.	Presents only if Soumya is unavailable.
Rahul Chainani	Application architecture, deployment, and integration facts.	Prepares answers; does not create a second presenter.
Pankaj Bharsakale	AI/RAG, model boundary, validator, and responsible-AI facts.	Prepares answers; does not create a second presenter.
Lakshmi Dudgikar	Requirements, cases, and user workflow mapping.	Prepares answers; does not create a second presenter.
Rahul Sharma	Testing, edge cases, and quality checks.	Prepares answers; does not create a second presenter.
Sweta Singh	Program statement, scope boundaries, and ROI framework.	Prepares answers; does not create a second presenter.

Safe two-case demonstration sequence

Time	Screen and action	What the presenter says
0:00–0:35	Case Queue: open Evidence for the documented closed case only.	This is a completed fictional outcome with recorded support. We are not changing or re-closing it.
0:35–1:15	Case Queue: open the default open/escalated case.	Status tells the workflow position; Rule(s) explain the signal; Action opens the context.
1:15–2:05	Investigation Workspace: run saved or calculated review.	The report is advisory. The validator and confidence card show support and limitations.
2:05–2:55	Evidence & RAG: show a source, exact chunks, FTS5/BM25 retrieval, and the linked record.	This is lexical retrieval over local fictional material, not a vector index and not proof.
2:55–3:40	Ask InvestigateIQ: ask one prepared bounded case question.	The answer points back to stored material and cannot submit a workflow action.
3:40–4:50	Show the human-decision and audit surfaces without clicking Submit.	Only a named person with a rationale can record Close, Request more information, or Escalate. We will not change a case in the live demo.

Demo safeguard

Do not submit an investigator or Compliance action, reset decisions, create intake data, or reveal or enter an Admin passcode during the live demonstration. The closed case is evidence-only; the open/escalated case is used to show the review path without persisting an action.

Final product position

InvestigateIQ demonstrates a transparent evidence-to-decision workflow. It helps a reviewer find, inspect, and question supplied material; it does not turn a record or model response into a legal, regulatory, or factual conclusion. The human reviewer remains responsible for the next step.

